Status	Finished
Started	Thursday, 29 October 2026, 6:54 PM
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Duration	4 mins 46 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the right sublist
c. At the beginning of the list
d. In the left sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It can only be applied to large lists
c. It is slower than linear search for small lists
d. It requires the list to be sorted ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Binary search
c. Linear search ✓
d. Hash search

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By dividing the list into halves
c. By comparing each element in the list sequentially ✓
d. By sorting the list first

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Must use a sorted array
c. There must be a mechanism to access middle element directly
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **6**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The middle element is chosen randomly
- b. The lower middle element is chosen as the middle element ✗
- c. The search stops
- d. The higher middle element is chosen as the middle element

Question **7**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. In the left sublist
- c. At the beginning of the list
- d. In the right sublist ✓

Question **8**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search continues in the right sublist
- b. The search continues in the left sublist
- c. The list is sorted
- d. The search ends successfully ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(log n)
- b. O(n log n)
- c. O(1)
- d. O👎 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Sorting the list before searching
- c. Dividing the list in half repeatedly
- d. Skipping every second element

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Hash search
- b. Binary search
- c. Linear search ✓
- d. Both (1) & (3)

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the list contains duplicate elements
- c. When the list is sorted
- d. When the element is found in the list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Inserting elements into a list
- c. Deleting elements from a list
- d. Sorting elements in a list

Question **14**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It always starts from the beginning of the list
- b. It can be applied only if the list is sorted ✓
- c. It works on unsorted lists
- d. It compares elements sequentially

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search requires the list to be sorted.
- c. Linear search divides the list into halves.
- d. Linear search is more efficient than binary search on large lists.